Machine Learning in Engineering Science

Week 13: Retrieval, source evidence, and structured knowledge

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Chapter 1

Retrieval, source evidence, and structured knowledge

Week 13: 2026-11-30, Monday 14:00–17:00.

Use original fictional lab passages and independently labeled questions. This lexical retrieval and extractive evidence workflow does not call a generative model; it is a baseline for, not a full implementation of, RAG.

Learning outcome

Build a lexical retrieval baseline; measure retrieval separately from grounded claim support; abstain without evidence.

The meeting contains three 60-minute modules: 60 minutes explanation and 120 minutes practice. The eight core exercises and two optional challenges are in-class practice, not additional weekly graded homework.

1.1 H1: Represent documents and retrieve evidence

1.1.1 Retrieval selects evidence, not truth

Find relevant passages before composing a source-bounded response.

Worked reasoning. Document identity; lexical vectors; cosine similarity; ranking; threshold; Task 1 and Task 2.

Check. Can the best-ranked passage be wrong?

Explanation and boundary. Yes. Ranking measures a chosen similarity, not the authority or factual correctness of a source.

1.1.2 Document identity must survive chunking

Store source ID, version and chunk location with every passage.

Worked reasoning. LAB-02:v1:chunk0 identifies an original teaching protocol passage; it is not an external scientific publication.

Check. What breaks when IDs are just row positions?

Explanation and boundary. Inserting a passage changes positional identities. Use stable source keys and preserve the indexed text snapshot.

1.1.3 Tokenization defines lexical overlap

A vocabulary maps terms to coordinates; preprocessing changes what matches.

Worked reasoning. Vocabulary [heat,limit,mean]; document 'heat heat limit' maps to [2,1,0].

Check. Does this representation preserve word order?

Explanation and boundary. No. Bag-of-words counts lose order; negation and relational meaning can be missed.

1.1.4 IDF discounts corpus-common terms

Smoothed IDF is ln((1+N)/(1+df))+1 in this notebook.

Worked reasoning. N=4, df=1 gives ln(5/2)+1=1.916; df=4 gives 1.0.

Check. Is a rare term necessarily an important fact?

Explanation and boundary. No. IDF is a statistical corpus weight. Rare typos and irrelevant words can also receive large weights.

1.1.5 TF-IDF is a fitted representation

Fit the vectorizer on the frozen corpus; transform questions with it.

Worked reasoning. The same query should be scored against one fixed index. New documents require an explicit index version.

Check. What happens to unseen query words?

Explanation and boundary. They do not create new fitted coordinates. A fully out-of-vocabulary query gives a zero vector under this baseline.

1.1.6 Cosine compares directions

Normalize vectors before using their dot product as similarity.

Worked reasoning. q=[1,1], d=[2,0]; cosine=2/(sqrt(2)*2)=0.707.

Check. Is cosine a probability that the passage is correct?

Explanation and boundary. No. It is a geometric similarity under the representation. A threshold is an operating rule, not calibrated confidence.

1.1.7 Top-k is a selection rule

Rank descending; break equal-score ties with stable source IDs.

Worked reasoning. If scores are equal, sort document IDs lexically so the same snapshot returns the same order.

Check. Can a deterministic tie rule improve relevance?

Explanation and boundary. Not inherently. It improves reproducibility and makes ambiguous rankings visible.

1.1.8 Thresholds trade coverage and abstention

Return no evidence when the maximum score is below a declared threshold.

Worked reasoning. Use an in-vocabulary protocol question and an unrelated vocabulary question. Inspect full score vectors.

Check. Does a low score prove the answer does not exist?

Explanation and boundary. No. Lexical mismatch can hide a relevant passage. Abstention is a limitation-aware action, not a proof of absence.

1.1.9 Chunking can remove required context

A passage without its units or exception can support the wrong claim.

Worked reasoning. Compare 'review if mean at least 80 C' with a chunk containing only 'review if mean at least 80'.

Check. Why is the shorter chunk risky?

Explanation and boundary. The missing unit changes interpretation, and retrieval alone cannot recover a qualifier absent from the supplied passage.

1.1.10 Summary: freeze the representation

Preserve corpus, source IDs, vectorizer configuration and ranking rule.

Worked reasoning. Show the exact quote, score and source version. Explain one lexical retrieval limitation.

Check. What will the next hour add?

Explanation and boundary. Independent relevance labels and claim-support checks. A retrieved citation is not yet a verified answer.

1.2 H2: Measure retrieval and check claim support

1.2.1 Two evaluation layers answer different questions

Retrieval asks whether evidence was found; grounding asks whether a claim follows.

Worked reasoning. A relevant passage can be retrieved while the answer misstates its threshold.

Check. Can perfect retrieval recall guarantee a correct response?

Explanation and boundary. No. Generation, interpretation and citation entailment can fail after retrieval succeeds.

1.2.2 Recall at k needs a relevance set

Recall@k = retrieved relevant sources / all labeled relevant sources.

Worked reasoning. Relevant={A,B}; top2={A,C}; recall@2=1/2. Precision@2=1/2.

Check. What if no relevant source is labeled?

Explanation and boundary. Do not divide by zero or fabricate relevance. Evaluate no-evidence/abstention cases separately.

1.2.3 Reciprocal rank rewards the first hit

For an answerable query, RR = 1/rank of the first relevant result.

Worked reasoning. [C,A,B] and [A,C,B] may have the same recall@3 but different reciprocal rank.

Check. Why report the rank convention?

Explanation and boundary. Metrics depend on k, relevance labels, tie rules and zero-hit handling. Preserve these definitions with the result.

1.2.4 An evidence quote has a narrow role

An exact substring proves text location, not whether every derived claim follows.

Worked reasoning. Source: threshold 80 C. Claim: threshold 95 C. A correct source ID does not rescue the wrong number.

Check. What does automatic quote containment miss?

Explanation and boundary. It misses paraphrase meaning, omitted exceptions and contradictions. Human or task-specific semantic checks remain necessary.

1.2.5 Task 3: score fixed retrieval questions

Use independently written relevance IDs; do not derive labels from ranking.

Worked reasoning. Compute macro recall only across answerable queries; list no-evidence outcomes separately.

Check. Why not label the top result as relevant?

Explanation and boundary. That would make the system grade itself and conceal retrieval failures.

1.2.6 Task 4: inspect false support

Test wrong citation IDs, fabricated quotes and changed numeric thresholds.

Worked reasoning. Unknown source; absent quote; quoted text with a contradictory threshold. They require different diagnoses.

Check. Will string checks catch every wrong claim?

Explanation and boundary. No. They are necessary structural checks in this exercise, not a universal entailment detector.

1.2.7 Task 5: change k under a fixed index

Keep questions, labels and threshold fixed while varying k.

Worked reasoning. Recall cannot decrease when adding ranked results under the same accepted query; precision can decrease.

Check. Why can a threshold obscure that comparison?

Explanation and boundary. A changed threshold changes which results are eligible. Keep it fixed and report abstention separately.

1.2.8 Treat source instructions as data

A malicious sentence in a passage cannot alter tool permissions.

Worked reasoning. The corpus includes an explicitly marked untrusted instruction. Retrieval may return it, but the workflow never executes it.

Check. Does a low similarity score secure a RAG system?

Explanation and boundary. No. Relevant malicious text can rank highly. Security enforcement must not depend on the retriever missing it.

1.2.9 Structured knowledge needs provenance too

A table row should retain units, source ID, version and extraction status.

Worked reasoning. name=review threshold; value=80; unit=C; source=LAB-02:v1; status=teaching fixture.

Check. Why not call this a validated engineering limit?

Explanation and boundary. It is an original fictional class protocol. It has no external safety authority or hardware validation.

1.2.10 Summary: evidence retrieval precedes explanation

Find a passage, preserve its identity, then check what it actually supports.

Worked reasoning. Save ranked IDs, scores, labels and one support failure. Do not export a bare average.

Check. What should a partner be able to reconstruct?

Explanation and boundary. Which question, index snapshot, ranking and relevance labels produced each reported metric.

1.3 H3: Retrieval studio: diagnose and package failures

1.3.1 Build an evidence-first retrieval record

Complete eight core tasks, then preserve an honest failure.

Worked reasoning. Restore 5 min; tasks 25 min; source audit 15 min; export and explain-back 15 min.

Check. What is the artifact, not just the score?

Explanation and boundary. A corpus snapshot, ranking configuration, per-query records, response fields, figures and hashes.

1.3.2 Task 1: distinguish answerable and unknown

Inspect the maximum cosine and abstention status for both questions.

Worked reasoning. The system returns no evidence instead of inventing a threshold or citation.

Check. What does the abstention support?

Explanation and boundary. Only that this index and operating rule supplied no accepted passage for this query.

1.3.3 Task 2: explain one weighted term

Compare term frequency, document frequency and IDF contribution.

Worked reasoning. An IDF of 1.0 is not zero; the default smoothed formula retains corpus-common terms.

Check. What changes if the corpus changes?

Explanation and boundary. Vocabulary, document frequencies, IDF, norms and rankings may all change. Preserve an index version.

1.3.4 Task 3: compute macro retrieval recall

Calculate per-query recall before averaging answerable cases.

Worked reasoning. Two queries with one and two relevant documents each receive equal weight under macro averaging.

Check. Is macro recall the same as pooled recall?

Explanation and boundary. Not generally. Pooled recall weights by the number of relevant documents. Name the aggregation you use.

1.3.5 Task 4: find a citation failure

Compare a truthful quote with a wrong-threshold claim.

Worked reasoning. Do not say only 'bad citation'. Identify the changed number, missing unit or absent quote.

Check. Can an automatic pass replace source reading?

Explanation and boundary. No. The class checks exact fields and narrow numeric facts; general semantic support remains a separate review.

1.3.6 Task 5: vary k without tuning on the final set

Use development questions for exploration; preserve held-out questions.

Worked reasoning. Record k and threshold before examining the held-out results. If already seen, label further work exploratory.

Check. What makes a final report optimistic?

Explanation and boundary. Repeatedly changing retrieval settings after seeing held-out outcomes and reporting only the best score.

1.3.7 Task 6: add a genuinely independent query

Write its relevant IDs before execution and justify them from source text.

Worked reasoning. Try a paraphrase to expose lexical mismatch. Explain whether a miss is due to wording or missing corpus evidence.

Check. Is one new query a representative benchmark?

Explanation and boundary. No. It is a diagnostic probe. Broader coverage requires a task-based question set.

1.3.8 Task 7: separate tool results from passages

A computed mean is not a retrieved protocol threshold.

Worked reasoning. Mean 81 C is computed from readings; threshold 80 C comes from the fictional protocol; comparison requests review.

Check. What remains outside this exercise?

Explanation and boundary. Physical measurement validity, real safety limits, automatic control authority and external deployment validation.

1.3.9 Task 8: export source-bounded evidence

Keep corpus hash and exact retrieved text with the metric table.

Worked reasoning. They select a query, locate its IDs and quotes, and reproduce its recall from the labeled relevant set.

Check. Why preserve text as well as a link?

Explanation and boundary. Sources and chunks can change. A frozen snapshot makes the reported computation inspectable.

1.3.10 Summary: ask what the source entails

Optional A changes chunking; optional B adds an adversarial passage.

Worked reasoning. Explain each outcome using a specific source or absent source, not a vague model-confidence story.

Check. What connects to the next week?

Explanation and boundary. The same untrusted evidence must pass through explicit permissions, budgets and approval gates before any action.

1.4 Mathematical reference

$$\mathrm{idf}(t) = \log \frac{1+N}{1+\mathrm{df}(t)} + 1, \qquad \cos(q,d) = \frac{q^\top d}{\|q\|\|d\|}.$$

$$\text{Recall@k} = \frac{|R \cap K|}{|R|}, \quad |R| > 0, \qquad \mathrm{RR} = \frac{1}{\text{first relevant rank}}.$$

Handle zero relevant sources as a separate no-evidence evaluation, not a zero-denominator recall. Lexical similarity is not calibrated confidence.

1.5 Eight core practice tasks

1.5.1 Task 1: Compare relevant evidence and abstention

Predict the returned source for a unit question and the behavior for an out-of-vocabulary question.

Self-check. The unknown query has no accepted passage. Similarity values are not calibrated factual-confidence probabilities.

1.5.2 Task 2: Inspect the fitted lexical representation

Predict whether a rarer word receives larger IDF. Then inspect actual vocabulary and normalized document coordinates.

Self-check. All six nonzero document vectors have L2 norm 1. Corpus-common terms still have nonzero smoothed IDF.

1.5.3 Task 3: Evaluate fixed held-out relevance labels

Freeze k=2 and threshold=0.12 before this cell. Predict a source for each held-out question and report no-evidence cases separately.

Self-check. Do not average an undefined recall for the no-evidence query as zero. After seeing this set, later tuning is exploratory.

1.5.4 Task 4: Separate citation structure from claim support

Predict which failure each answer should trigger. Exact-quote checking is only one layer of support.

Self-check. Even the first answer still requires semantic review. The checker handles a narrow numeric fixture, not general entailment.

1.5.5 Task 5: Change k on development questions only

Compare k=1 and k=2 with the same fitted corpus and threshold. Keep held-out results fixed and name this as development exploration.

Self-check. Recall cannot decrease when an unchanged accepted ranked list is extended. This is not permission to tune on already inspected held-out outcomes.

1.5.6 Task 6: Add one independently labeled diagnostic query

Write a paraphrase and relevant IDs before retrieval. The supplied example is a diagnostic, not a representative new benchmark.

Self-check. A lexical miss is useful evidence. Do not silently relabel a missed source as irrelevant after seeing the ranking.

1.5.7 Task 7: Combine a numeric result and a source rule

Compute the mean independently, retrieve the class threshold passage and keep their provenance separate.

Self-check. The value 81 is computed, while the threshold 80 is source-derived. Neither establishes a real equipment limit.

1.5.8 Task 8: Preserve corpus identity and one failure

Export ranked IDs, exact text, fixed labels and one unsupported claim. The source snapshot is part of the evidence.

Self-check. Source IDs and exact quotes are provenance anchors, not a guarantee that the source is true.

1.6 Evidence handover and summary

Build a lexical retrieval baseline; measure retrieval separately from grounded claim support; abstain without evidence. Save eight predictions, eight explanations and three reflections. Preserve source, data, versions, traces and one limitation. The notebook exports a draft when fields are blank;

response completeness does not establish correctness. Save the edited notebook separately if changing code.

1.7 Primary references

- Retrieval-Augmented Generation: https://arxiv.org/abs/2005.11401.
- TfidfVectorizer reference: https://scikit-learn.org/stable/modules/generated/sklearn.feature_extraction.text.TfidfVectorizer.html.